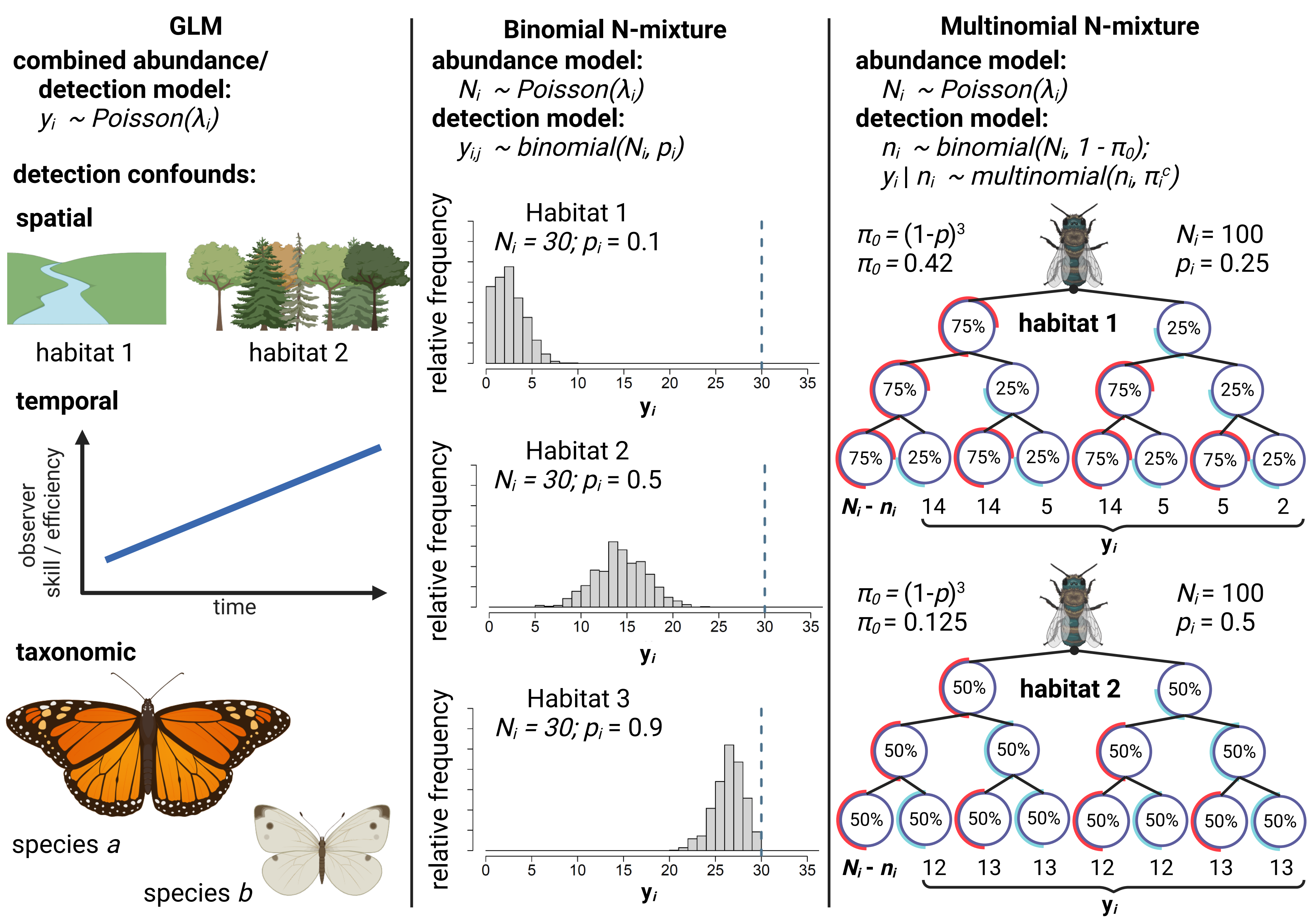
GLM
combined abundance/
detection model:
$y_i \sim Poisson(\lambda_i)$
detection confounds:
spatial
habitat 1
habitat 2
temporal
observer skill / efficiency
time
taxonomic
species *a*
species *b*
Binomial N-mixture
abundance model:
$N_i \sim Poisson(\lambda_i)$
detection model:
$y_{i,j} \sim binomial(N_i, p_i)$
Habitat 1
$N_i = 30; p_i = 0.1$
Habitat 2
$N_i = 30; p_i = 0.5$
Habitat 3
$N_i = 30; p_i = 0.9$
relative frequency
0 5 10 15 20 25 30 35
y_i
Multinomial N-mixture
abundance model:
$N_i \sim Poisson(\lambda_i)$
detection model:
$n_i \sim binomial(N_i, 1 - \pi_0);$
$y_i \mid n_i \sim multinomial(n_i, \pi_i^c)$
$\pi_0 = (1-p)^3$
$\pi_0 = 0.42$
$N_i = 100$
$p_i = 0.25$
habitat 1
75% 25%
$N_i - n_i$ 14 14 5 14 5 5 2
y_i
$\pi_0 = (1-p)^3$
$\pi_0 = 0.125$
$N_i = 100$
$p_i = 0.5$
habitat 2
50%
$N_i - n_i$ 12 13 13 12 12 13 13
y_i